AI4S OPEN INNOVATION · AI FOR LIFE SCIENCE · FINAL ROUND

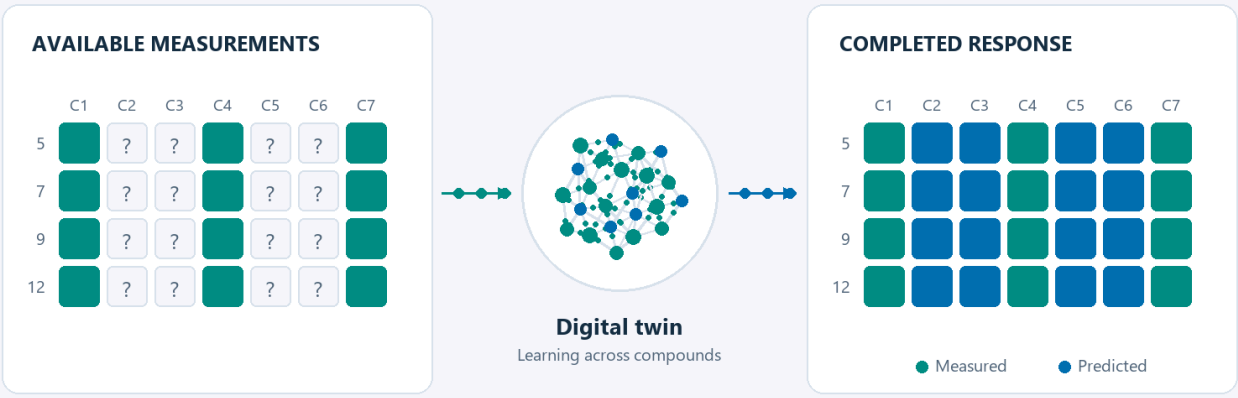
NeuroChip Twin V2

A few-shot, uncertainty-aware digital twin for neural organ-on-a-chip neurotoxicity screening

Francisco Angulo de Lafuente · Team 19 · 30 October 2026

From three concentrations to a complete response

The context includes measurements at the time points covered by the study.



Predicts responses at unmeasured concentrations and reports uncertainty.

CONCEPTUAL DIAGRAM · NO EXPERIMENTAL VALUES

Method: NeuroTrajectory / technical report

−10.96 %
mean absolute error vs the strongest of five baselines
243 unseen compounds

89.9 %
observed coverage of 90 % prediction intervals
calibrated uncertainty

86.4 %
balanced accuracy of the hazard call
from only three concentrations

57.1 %
fewer treated wells
three of seven concentrations

EVERYTHING IS PUBLIC AND READY TO REPLICATE

All experiments, code, trained models, results and papers are in our public GitHub repository and in our Kaggle notebook and dataset. The answer to the Review Committee's question is on page 2.

github.com/Agnuxo1/neurochip-twin · kaggle.com/code/franciscoangulo/neurochip-twin-v2-demo

The Review Committee's question

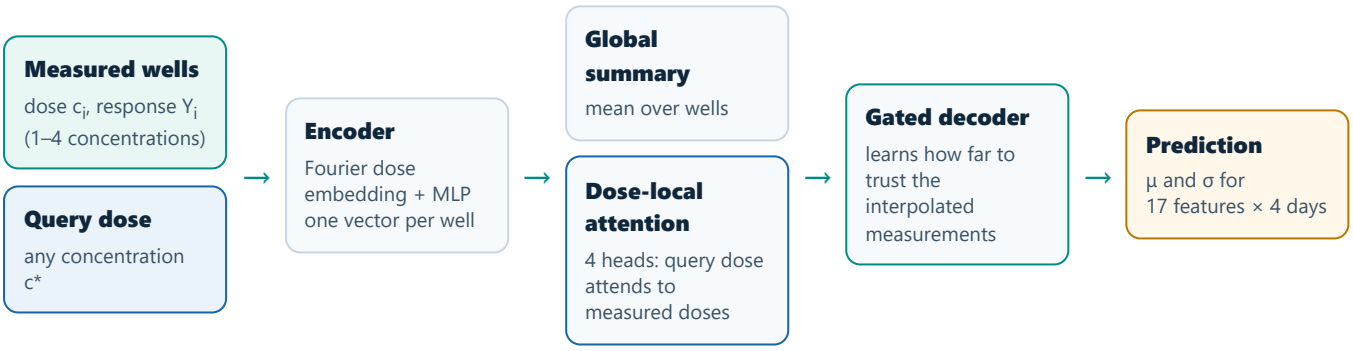
QUESTION FROM THE REVIEW COMMITTEE · FINAL ROUND, 30 OCTOBER 2026

"Is your model based on a regression neural network?"

ANSWER

Yes, in the sense that it is a regression network: it predicts **continuous values** (the full response at each concentration and time point) **together with its uncertainty**. It is **not a recurrent network** (no RNN or LSTM). It is a **conditional neural process**: an **encoder-decoder with attention over the measured doses**, **meta-trained across 243 substances**, so that it can **generalise to a new substance from only three measured concentrations**.

HOW THE MODEL WORKS



Task	**Probabilistic regression**: a mean and a standard deviation for every output
Output	**All 17 network features × 4 developmental days**, at any concentration, measured or not
Input	As few as **1 to 4 measured concentrations** of a new substance (its wells)
Architecture	**Conditional neural process**: shared encoder, global summary + **dose-local multi-head attention**, **interpolation-informed gated decoder**
Training	**Meta-learned across 243 substances** (Gaussian likelihood), split by substance for evaluation
Not used	No recurrence (no RNN / LSTM): the time points are predicted **jointly**, as one trajectory

Background

The project builds on an idea from my earlier paper, **"Speaking to Silicon: Neural Communication with Bitcoin Mining ASICs"** (arXiv:2601.12032): **using a partial signal to decide whether a costly process should continue**. In NeuroChip Twin this idea underlies the investigation of **early stopping in biological assays**.

Reproducibility

All code, results and the technical report are public at github.com/Agnuxo1/neurochip-twin. Should any step fail to reproduce in your environment, please let me know and I will correct it promptly.

At a glance

- **A digital twin trained and validated on real electrophysiology**: 243 compounds and 6,902 wells from the public US EPA Network Formation Assay.
- **Few-shot by design**: from as few as one to four measured concentrations, it forecasts the full concentration × developmental-day response of a compound it has never seen.
- **Clear, measured gain**: **−10.96 % error** against the strongest of five baselines, with a paired 95 % confidence interval that excludes zero.
- **Trustworthy by construction**: calibrated intervals with **89.9 % coverage** at a 90 % target, across cytotoxic and non-cytotoxic compounds.
- **Built for the laboratory**: hazard call, next-dose selection and early-stopping analysis, plus a directed layer fitted on a **real four-compartment neural chip**.
- **Fewer wells for the same answer**: three of seven concentrations means **57.1 % fewer treated wells** in a standard design.
- **Fully open and reproducible**: MIT licence, CPU demo in under a minute, every reported number generated from a result file and checked automatically.

SCALE OF THE WORK

18
documented experiments, each with its own script and result file

243
compounds · 6,902 wells · 17 features · 4 days

15
trained models (5 folds × 3 seeds)

106
automated tests, continuous integration

440
axons from 21 real chip recordings

< 1 min
CPU demo, no GPU or internet needed

EVIDENCE AT A GLANCE

Forecast unseen compounds	**−10.96 %** error vs best baseline; better for **77 %** of compounds
Every measurement budget	Lower error with **1, 2, 3 and 4** measured concentrations
Calibrated uncertainty	**80.0 / 89.9 / 94.9 %** coverage at 80 / 90 / 95 % targets
Hazard call from three doses	**86.4 %** balanced accuracy vs 82.6–85.9 % for the EPA rules (same 97 compounds)
Exact reproduction of EPA rules	**7 of 7** reference rules, 0.0 pp difference
Each component matters	Ablation: **+6.9 %** error without the interpolation-informed decoder
Next-dose selection	**9.1 %** better than random; activity accuracy **91.0 %** (anchored)
Real neural chip layer	**440** axons; **9.0 %** lower feed-forward error, $p < 0.001$
Cross-assay corroboration	**95.8 %** of morphology-active compounds also active in the network assay

Animated versions of every figure are **attached to this PDF** (attachments / paperclip panel) and appear in the presentation video.

Built on published research

PUBLISHED ON ARXIV · JANUARY 2026

Speaking to Silicon: Neural Communication with Bitcoin Mining ASICs

F. Angulo de Lafuente, V. Veselov, R. Goodman · arXiv:2601.12032 · DOI 10.13140/RG.2.2.17077.33765

The paper shows that **early, partial information carries predictive power** about a costly process, and uses it to decide **whether that process should continue** (the Thermodynamic Probability Filter), with **machine-checked proofs in Lean 4**.

NeuroChip Twin V2 carries the same principle into biology: a few measured concentrations are enough to forecast the rest of the experiment, and the same idea drives our investigation of **early stopping in neural assays**.

From prototype to real data

NeuroChip Twin v1 (public, tag *v1-synthetic*) established the temporal-phenotyping pipeline on synthetic data. **Version 2 is trained and validated on real biological recordings**, with a frozen, provenance-tracked dataset and a new validation framework.

Team profile: AI + biology

Francisco Angulo de Lafuente is the **sole inventor of two Spanish patent families on microbial bioconversion** (CPC C12P, fermentation processes: ES2273594B1, ES2341194B1, PCT WO2009034217A1), co-inventor of two further process patents, author of the book *ECOFA* (ISBN 978-84-936176-4-6), of **four arXiv preprints (2026)** and of more than 150 public repositories with original neuromorphic and reservoir-computing architectures.

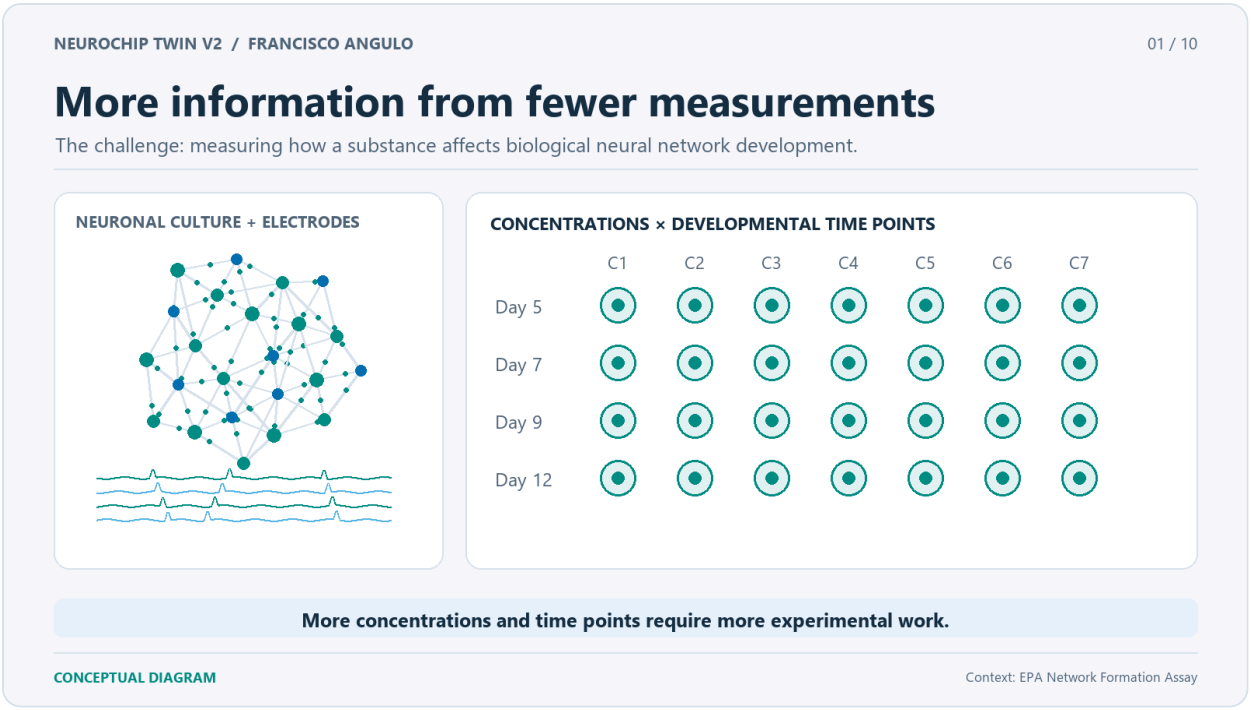
EVERYTHING IS PUBLIC AND READY TO REPLICATE

Experiments, code, trained models, result files, technical report and papers are in our public GitHub repository, mirrored in our Kaggle notebook and dataset.

GitHub repository	github.com/Agnuxo1/neurochip-twin
Kaggle notebook (CPU, no internet)	www.kaggle.com/code/franciscoangulo/neurochip-twin-v2-demo
Kaggle dataset (code, models, results)	www.kaggle.com/datasets/franciscoangulo/neurochip-twin-v2
Technical report (PDF)	github.com/Agnuxo1/neurochip-twin/releases/download/v2.0.0/neurochip_twin_v2.pdf
Interactive demo	agnuxo1.github.io/neurochip-twin/
Presentation video	youtu.be/yLzpvGqP_2A
Earlier paper: Speaking to Silicon	arxiv.org/abs/2601.12032

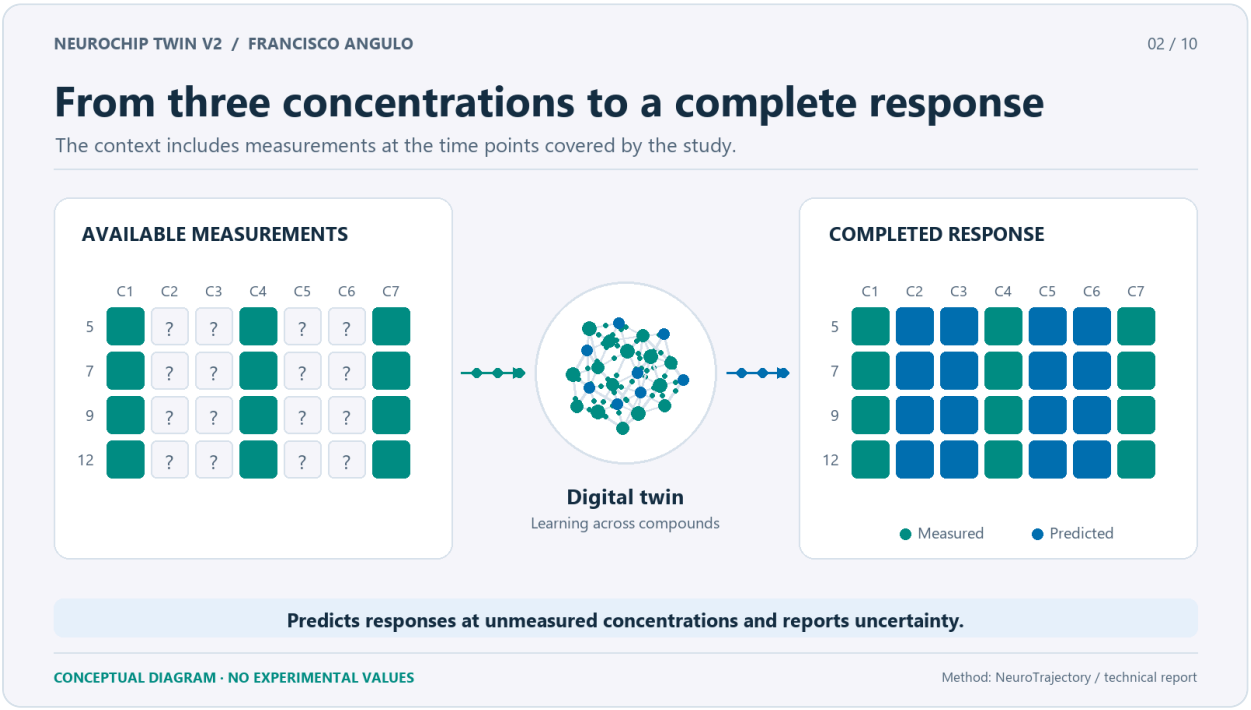
The problem and the method

01 More information from fewer measurements



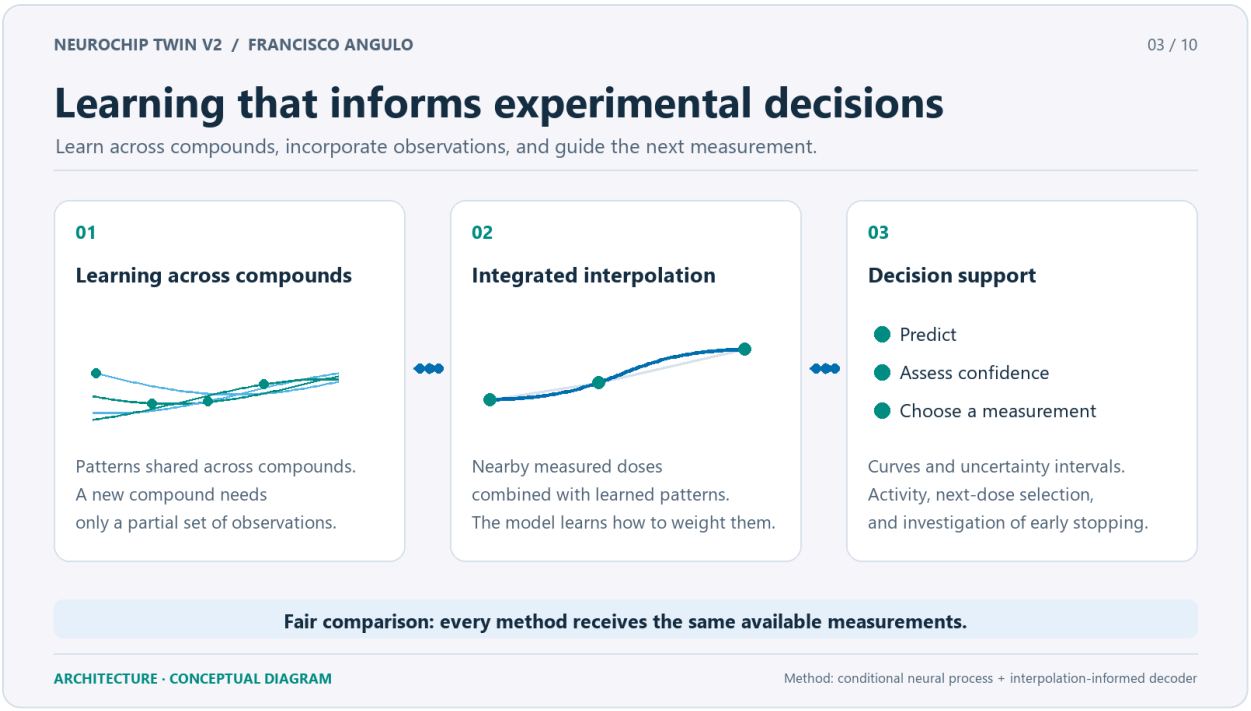
Developmental neurotoxicity screening records network activity for days at up to seven concentrations. **NeuroChip Twin V2 turns a few measured concentrations into the complete concentration × day response.**

02 From three concentrations to a complete response



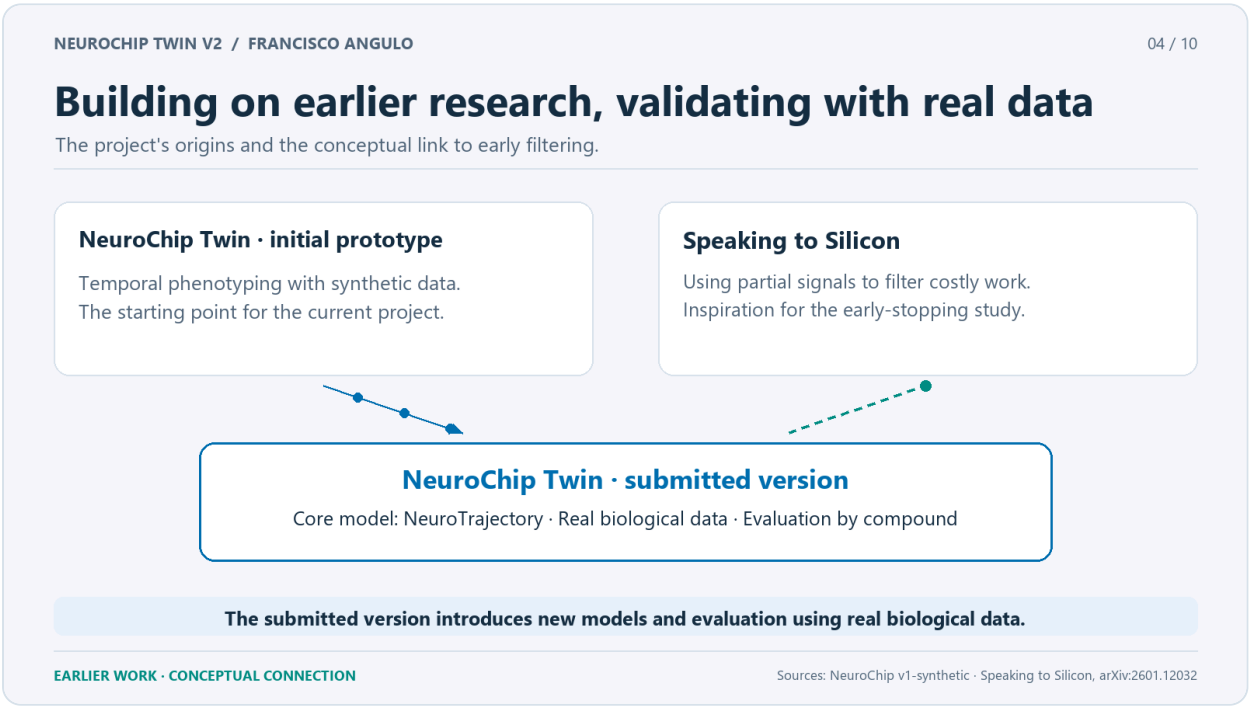
One model, meta-learned across 243 compounds, forecasts all 17 network features at days 5, 7, 9 and 12 for a compound it has never seen, **with a calibrated uncertainty interval for every prediction.**

03 Learning that informs experimental decisions



Three contributions: **learning across compounds**, an **interpolation-informed gated decoder with dose-local attention**, and a **decision layer** for the laboratory. Ablations confirm that each component contributes: removing the interpolation-informed decoder raises the error by **6.9 %**.

04 Building on earlier research, validating with real data



From the synthetic v1 prototype to **real biological data and a new validation framework**. The early-stopping concept comes from our published work **"Speaking to Silicon" (arXiv:2601.12032)**.

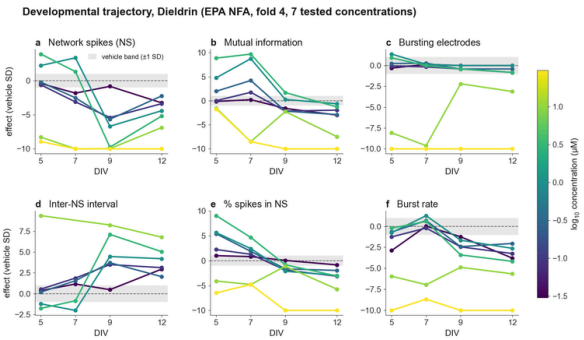
Real data and rigorous evaluation

05 Real biological data

NEUROCHIP TWIN V2 / FRANCISCO ANGULO 05 / 10

Real biological data

EPA Network Formation Assay: rat neuronal cultures with electrical readouts.



243
compounds

17
neural activity features

4
time points: days 5, 7, 9, 12

One compound is shown here; the main performance result is evaluated across compounds.

REAL DATA · ORIGINAL REPOSITORY FIGURE

Source: EPA NFA · scripts/make_visuals.py · Dieldrin example

243 compounds, 6,902 wells, 17 network features and 4 developmental days from the public US EPA Network Formation Assay. **All 7 EPA reference hit-call rules are reproduced exactly** (0.0 percentage-point difference from the published values).

06 Evaluation on compounds unseen during training

NEUROCHIP TWIN V2 / FRANCISCO ANGULO 06 / 10

Evaluation on compounds unseen during training

Five compound-level folds: each test compound is excluded from its models' training data.

ROTATING THE OUTER TEST FOLD

	Fold 0	Fold 1	Fold 2	Fold 3	Fold 4
Round 1	Test	Train	Train	Train	Train
Round 2	Train	Test	Train	Train	Train
Round 3	Train	Train	Test	Train	Train
Round 4	Train	Train	Train	Test	Train
Round 5	Train	Train	Train	Train	Test

Additional robustness check

Exclude fold 0, which was inspected during development.

194
compounds in folds 1–4

−11.25%
error relative to the baseline

All samples from each compound remain in the same fold.

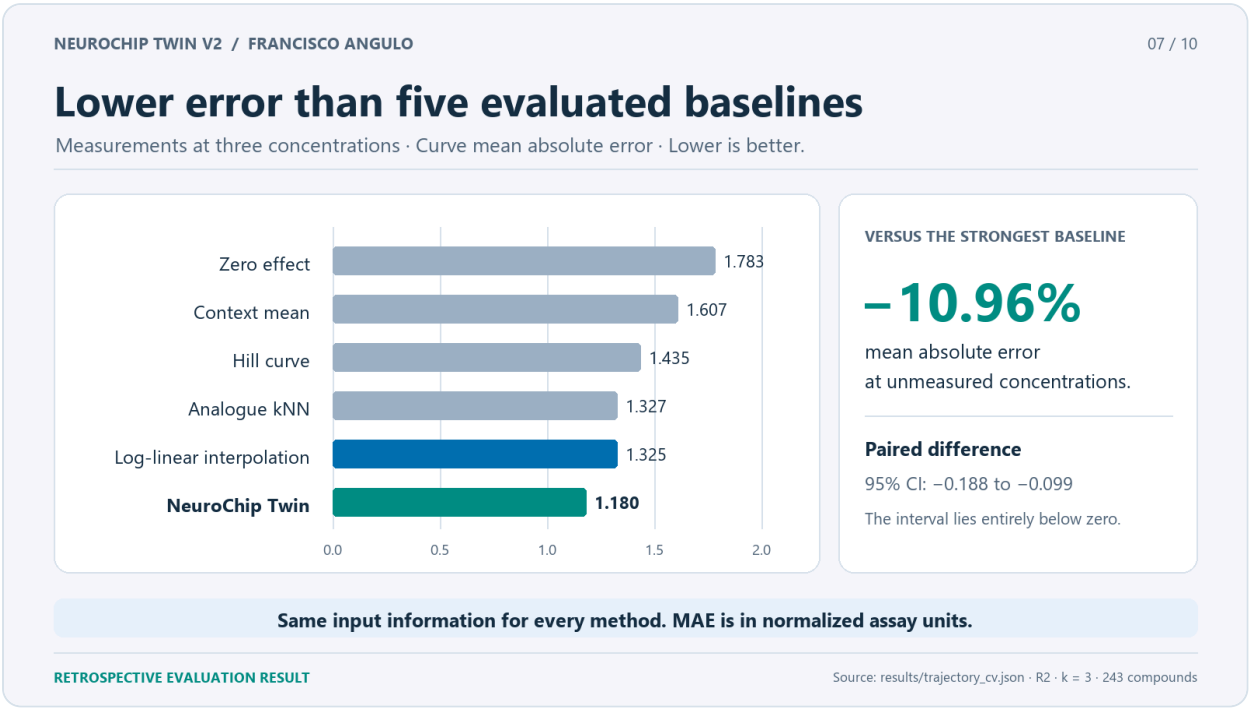
EVALUATION PROTOCOL · SCHEMATIC VIEW

Source: results/trajectory_cv.json · compound-level cross-validation

5-fold cross-validation split by compound, 3 training seeds and 5 dose designs per budget. The gain holds on the folds never used during development: **−11.25 %** error with three concentrations.

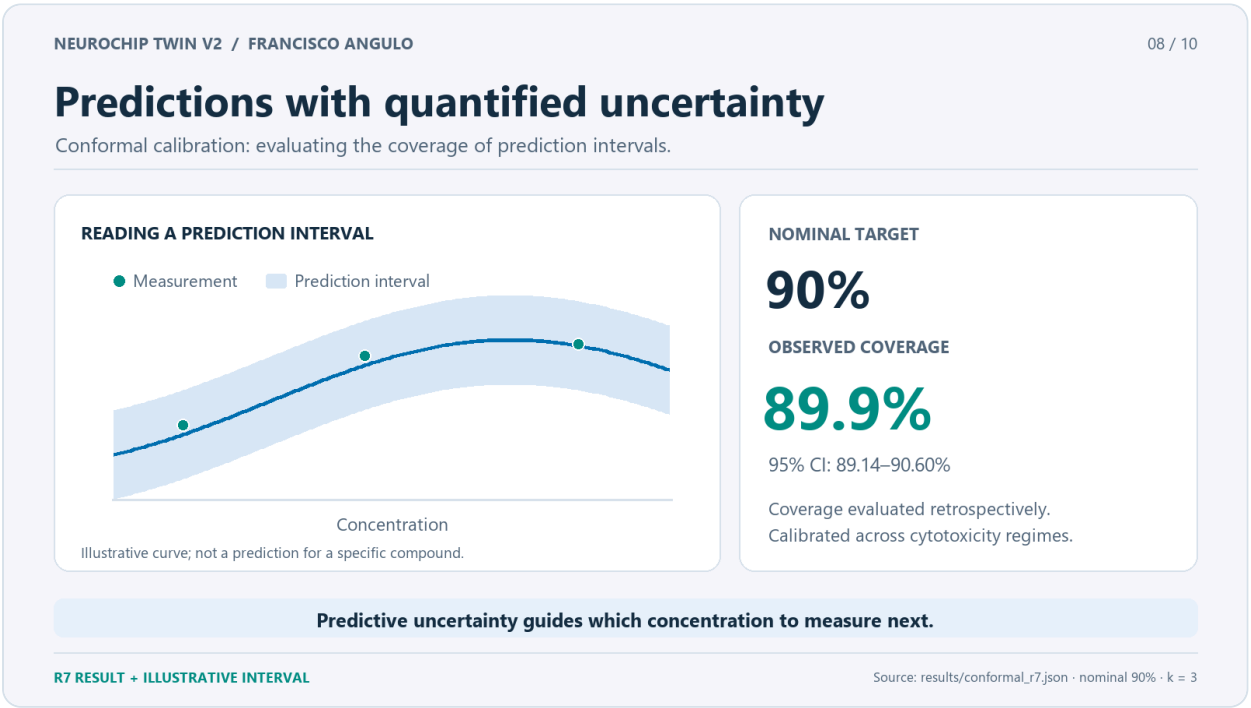
Results

07 Lower error than five evaluated baselines



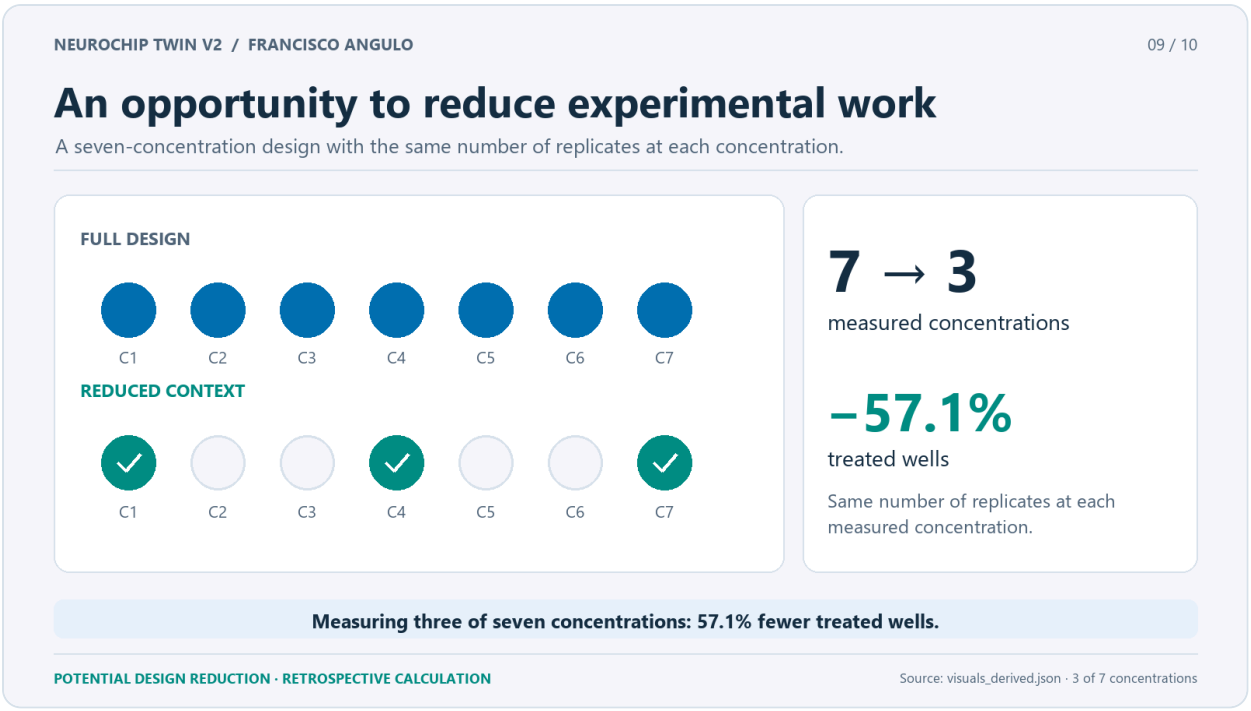
−10.96 % mean absolute error against the strongest baseline, log-linear interpolation (1.180 vs 1.325; paired 95 % CI -0.188 to -0.099), **better for 77 % of the compounds**; −17.8 % vs Hill curves, −11.1 % vs analogue kNN and −26.6 % vs the context mean. **Lower error at every measurement budget**: −5.3 %, −8.2 %, −11.0 %, −8.1 % with 1, 2, 3 and 4 concentrations.

08 Predictions with quantified uncertainty



89.9 % observed coverage for 90 % intervals (95 % CI 89.1–90.6 %), and **80.0 % / 94.9 %** for 80 % / 95 % targets. Calibration holds for cytotoxic (90.0 %), non-cytotoxic (89.7 %) and indeterminate compounds (90.5 %).

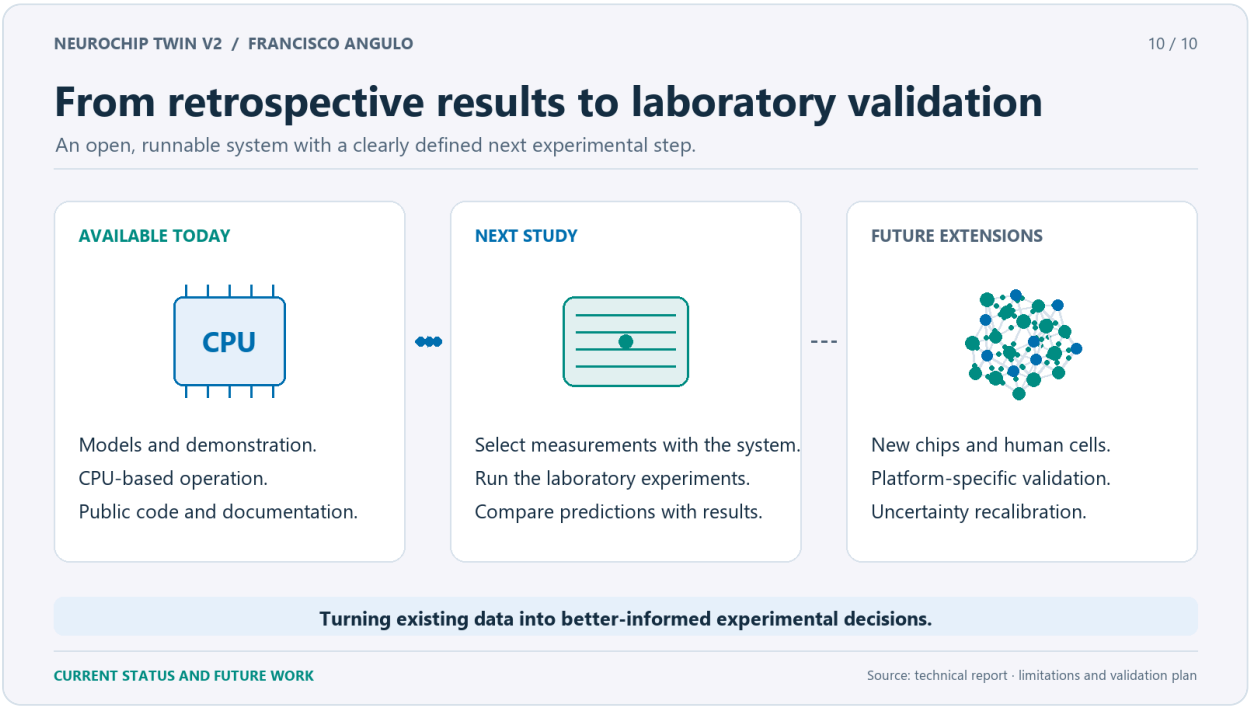
09 An opportunity to reduce experimental work



Measuring **3 of 7 concentrations means 57.1 % fewer treated wells** (71.4 % with two). From the same three concentrations, the hazard call reaches **86.4 % balanced accuracy**, above the 82.6–85.9 % of the three EPA reference rules on the same 97 compounds.

From model to laboratory

10 From retrospective results to laboratory validation



Ready to use today: trained models, a CPU demo that runs in under a minute, an interactive app and full documentation. **DoseCompass** proposes the next concentration to measure (**9.1 % better than random**); its anchored variant raises activity-call accuracy to **91.0 %** (88.1 % with fixed log-spaced doses).

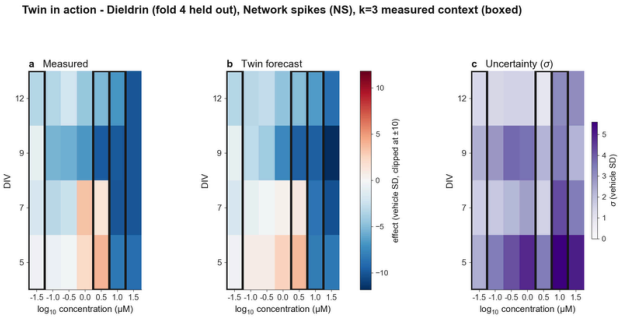
11 A real example: measurements, prediction, uncertainty

NEUROCHIP TWIN V2 / FRANCISCO ANGULO

Q&A SUPPORT

A real example: measurements, prediction, uncertainty

Dieldrin · Held out in fold 4 and excluded from its models' training data.



01 Real measurements

The outlined columns are the context concentrations.

02 Twin prediction

Predicts unmeasured concentrations. The model was not trained on Dieldrin.

03 Uncertainty

The figure shows the model's predictive uncertainty scale.

Representative example; improvement and coverage figures come from the aggregate evaluation.

REAL PREDICTIONS AND MEASUREMENTS · ORIGINAL FIGURE

Source: twin_in_action_16x9.png · EPA NFA · k = 3

Dieldrin, a compound held out from training: **the measured concentrations, the twin's forecast for the unmeasured ones and its uncertainty**, side by side.

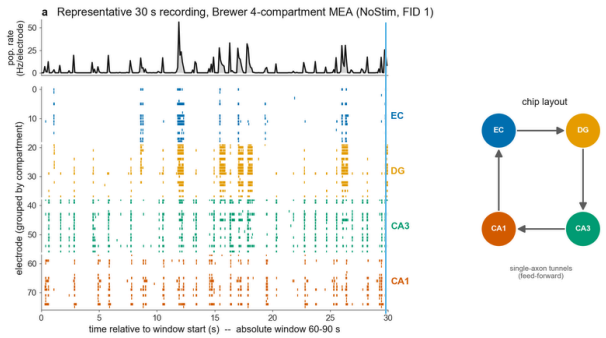
12 Activity recorded from a real neural chip

NEUROCHIP TWIN V2 / FRANCISCO ANGULO

Q&A SUPPORT

Activity recorded from a real neural chip

Brewer data: a four-compartment hippocampal system with multielectrode readouts.



76
electrodes in this recording

6,921
spike events in 30 seconds

Axon-resolved recordings feed the directed chip layer of the twin.

EPA data evaluate compound-response prediction; these recordings characterize the neural chip.

REAL BIOLOGICAL RECORDING · DESCRIPTIVE ANALYSIS

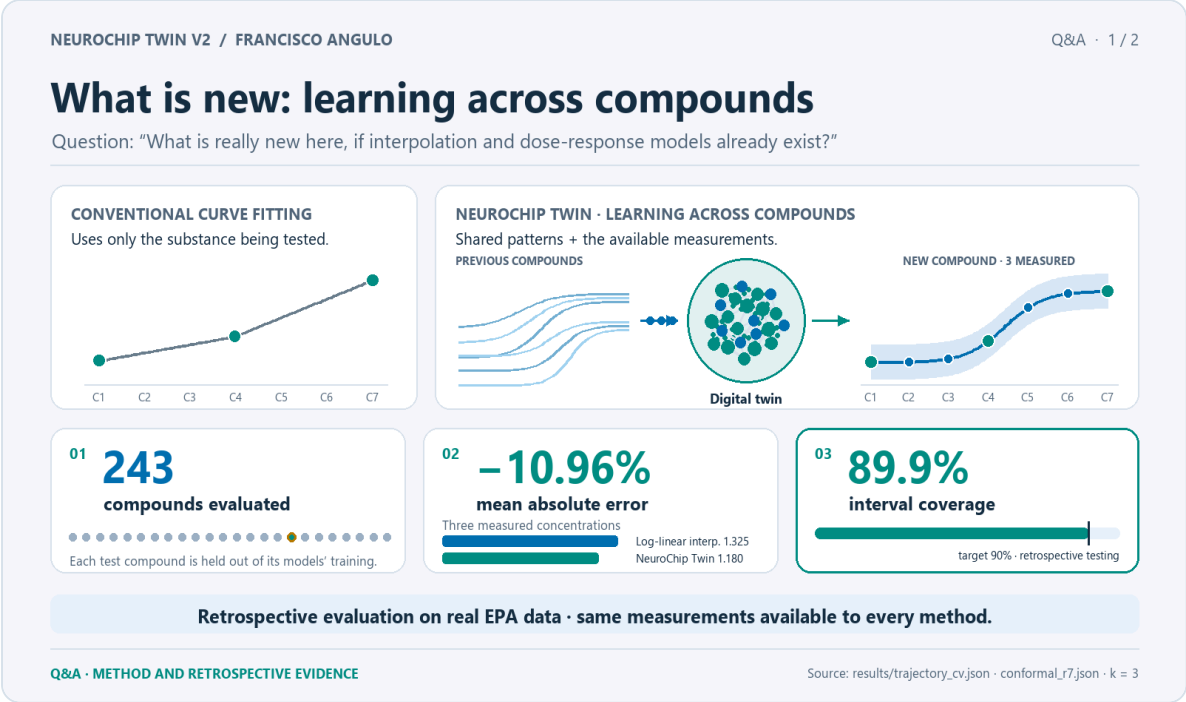
Source: real_network_raster_16x9.png · NoStim · FID 1 · 60–90 s window

A directed chip layer fitted on **440 axons from 21 recordings of a real four-compartment hippocampal chip**: **9.0 % lower feed-forward-fraction error** than the tunnel-mean baseline (bootstrap, 10,000 resamples, $p < 0.001$).

Prepared answers from the presentation (1/2)

QUESTION 1 · ANSWERED IN THE NARRATED PRESENTATION

"What is really new here, if interpolation and dose-response models already exist?"



ANSWER (AS NARRATED IN THE FINAL)

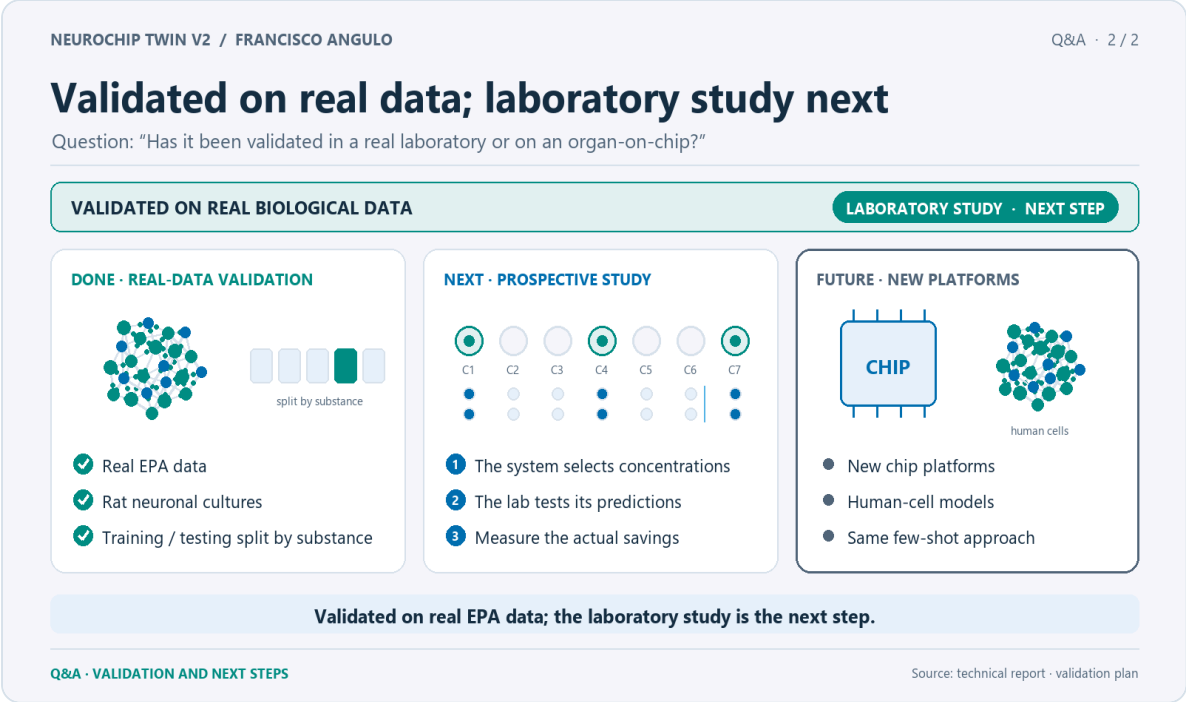
Conventional interpolation and curve fitting mainly use measurements from the substance being tested. NeuroChip Twin also **learns shared patterns across compounds**, and **learns how to combine these patterns with the available measurements**. We evaluated it on **a dataset of 243 substances**, keeping each test compound out of its models' training data. **With measurements at three concentrations, mean absolute error fell by about 11% against the strongest evaluated baseline**. Its calibrated prediction intervals achieved **89.9% coverage for a 90% target** in retrospective testing.

Evidence: results/trajectory_cv.json (243 compounds, compound-level cross-validation) and results/conformal_r7.json. The animated version of this card is attached to the PDF.

Prepared answers from the presentation (2/2)

QUESTION 2 · ANSWERED IN THE NARRATED PRESENTATION

“Has it been validated in a real laboratory or on an organ-on-chip?”



ANSWER

The current validation uses **real EPA data from rat neuronal cultures**, with **training and testing split by substance**. The next step is **a prospective study with a laboratory**: **the system chooses which concentrations to measure, and the lab tests its predictions**. That study will **measure the actual savings**.

Evidence: results/trajectory_cv.json (training and testing split by substance) and the laboratory study plan in the technical report. The animated version of this card is attached to the PDF.

Replicate it in one minute

EVERYTHING IS PUBLIC AND READY TO REPLICATE

Everything — experiments, code, models, results and papers — is public on GitHub and Kaggle, under the MIT licence.

```
pip install -r requirements.txt
python demo.py          # CPU only, no internet, under one minute
```

Each of the **18 experiments** has its own script and result file; **every number in the report is generated from those files and verified automatically** (*scripts/check_numbers.py*), and the model bundle is checked against its **SHA-256 manifest**.

GitHub repository	github.com/Agnuxo1/neurochip-twin
Kaggle notebook (CPU, no internet)	www.kaggle.com/code/franciscoangulo/neurochip-twin-v2-demo
Kaggle dataset (code, models, results)	www.kaggle.com/datasets/franciscoangulo/neurochip-twin-v2
Technical report (PDF)	github.com/Agnuxo1/neurochip-twin/releases/download/v2.0.0/neurochip_twin_v2.pdf
Interactive demo	agnuxo1.github.io/neurochip-twin/
Presentation video	youtu.be/yLzpvGqP_2A
Earlier paper: Speaking to Silicon	arxiv.org/abs/2601.12032

EARLIER PUBLISHED WORK

Speaking to Silicon: Neural Communication with Bitcoin Mining ASICs

F. Angulo de Lafuente, V. Veselov, R. Goodman · arXiv:2601.12032